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A Method of Analysis of Vibration–Rotational and Rotational Spectra of Light Diatomic Molecules

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The potential model of G. Thompson *et al.* [*J. Mol. Spectrosc.* **124,** 130–138 (1987)] for an analysis of the vibrational–rotational and the rotational spectra of diatomic molecules is modified by an algebraic WKB treatment of the Schrödinger equation given by Watson's effective Hamiltonian. A compact expression is given for the contributions from the adiabatic and nonadiabatic correction terms in the Hamiltonian to the Dunham's Y_{ij} coefficients. The present model is applied to the reported vibrational–rotational and rotational spectra of LiH as a typical light diatomic molecule where effects of the diabatic and nonadiabatic corrections are significant, and all spectral lines of all isotopomers are fitted to a single set of molecular constants simultaneously well within experimental errors. The analytical expressions for $\Delta_\omega^{a,b}$ and $\Delta_B^{a,b}$ obtained in the present study provide the basis for the validity of the potential model of Thompson *et al.* for molecules composed of heavier atoms. © 1997 Academic Press

INTRODUCTION

High-resolution vibrational–rotational spectra of diatomic molecules with infrared diode lasers or Fourier transform spectrometers provide many spectral lines for many vibrational states of various isotopomers. They require consideration of adiabatic and nonadiabatic corrections (*1–6*) to the Born–Oppenheimer approximation to analyze those vibrational–rotational and rotational transitions, generally observed in microwave or millimeter-wave spectra.

A conventional method of analysis was to fit the spectral lines to the (v, J) levels of a molecule ab expressed in

$$E(v, J) = \sum_{i,j} \mu^{-(i+2j)/2} U_{ij} [1 + (m_e/M_a)\, \Delta_{ij}^{a} + (m_e/M_b)\, \Delta_{ij}^{b}](v + 1/2)^i [J(J+1)]^j, \quad [1]$$

which was given by Bunker (*1*) and was justified by Watson (*2*), in which U_{ij} and $\Delta_{ij}^{a,b}$ are the mass-invariant constants. The empirical parameters U_{ij} and $\Delta_{ij}^{a,b}$, originally introduced empirically by Ross *et al.* (*7*) to take the Born–Oppenheimer corrections into account, were considered theoretically by Watson (*2, 3*). However, when a least-squares fit of the spectral lines was made using Eq. [1] all parameters U_{ij} and Δ_{ij} were commonly treated as freely floating parameters in conflict with Watson's theory (*8*). If all U_{ij} and Δ_{ij} float freely, one cannot necessarily expect that these parameters determined reflect correctly (*4, 10*) Dunham's potential energy (*9*). Watson (*2*) described the importance of conditions that mutually constrain the U_{ij} parameters.

Thompson *et al.* (*11*) proposed a model for the fit to take into account these constraints. They introduced empirical parameters Δ_ω and Δ_B for ω_e and B_e, similar to Watson's Δ_{10} and Δ_{01} terms, respectively, and analyzed the vibrational–rotational and rotational lines in a least-squares fit to the Dunham potential energy parameters ω_e, B_e, a_1, a_2, a_3, . . . , and other Δ_ω and Δ_B. We analyzed the spectra of BrCl (*12*), GeS (*13*), and GeSe (*14*) with that model, which worked well for these massive molecules.

However, Maki *et al.* (*15*) reported that their model (*11*) did not work well for light molecules such as LiH, NaH, HF, and CO. After this paper, Ogilvie (*8, 16*) reported his success in overcoming the difficulty of using parameters for a radial potential function in which the Born–Oppenheimer correction terms were included. His initial attempt to fit the LiH spectra did not include nonadiabatic vibrational effects, but he later used a full effective Hamiltonian which included all adiabatic and nonadiabatic effects (*5, 17*).

Ogilvie (*5*) represented Δ_{ij} in terms of the adiabatic, nonadiabatic vibrational, and nonadiabatic rotational effects in a perturbation treatment (*17*). These three effects were expressed as a Taylor expansion of a normalized internuclear distance z $[=2(r - r_e)/(r + r_e)]$ with coefficients denoted by $s_j^{a,b}$, $t_j^{a,b}$, and $u_j^{a,b}$, respectively. In his analysis of vibrational–rotational and rotational spectra, the wavenumbers were fit to his potential model, retaining $s_j^{a,b}$, $t_j^{a,b}$, and $u_j^{a,b}$ as adjustable parameters as well as the potential energy coefficients c_j (*5*) or with constraints from theoretical results (*18*).

An approach different from such a direct potential fit as above was reported by Tiemann and Ogilvie (*4*) and Ogilvie (*19*), who proposed a systematic method for fitting the spec-

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tra to the empirical parameters U_{ij} and Δ_{ij} in Eq. [1]. They considered Δ_{ij} as purely empirical and did not constrain Δ_{ij} because some were poorly determinable, but they made full use of interrelations between U_{ij} through the Dunham potential energy coefficients a_j. Their method was later applied to an analysis of rotational spectra of lithium hydride measured with a tunable far-infrared laser spectrometer (*20*).

Although the potential model of Thompson *et al.* (*11*) works well for many heavier molecules, for light molecules it needs to be modified algebraically by the introduction of expressions for $\Delta_\omega^{a,b}$ and $\Delta_B^{a,b}$ and additional parameters representing the contributions of the adiabatic and nonadiabatic effects. The purpose of the present paper is to modify this model to extend its application to light molecules such as LiH.

EFFECTIVE RADIAL EQUATION

As the method of Thompson *et al.* (*11*) is based on Watson's model, we start from the effective Hamiltonian given by Watson (*2*) for a rotating and vibrating diatomic molecule. Watson derived from the effective Hamiltonian a versatile equation [1] with empirical parameters Δ_{ij}. However, we solve the Schrödinger equation algebraically because analytic expressions for Δ_ω and Δ_B and for additional parameters must be derived.

With an effective Hamiltonian, the Schrödinger equation is written in terms of a variable $\xi = (r - r_e)/r_e$ as

$$\left[-\frac{h^2}{8\pi^2\mu_c r_e^2}\frac{d}{d\xi}\{1 + (m_e/M_a)Q_a(\xi) + (m_e/M_b)Q_b(\xi)\}\frac{d}{d\xi} + \frac{h^2}{8\pi^2\mu_c r_e^2(1+\xi)^2}\{1 + (m_e/M_a)R_a(\xi) + (m_e/M_b)R_b(\xi)\}J(J+1) + V(\xi) + (m_e/M_a)S_a(\xi) + (m_e/M_b)S_b(\xi)\right]\Psi_{vJ}(\xi) = E_{vJ}\Psi_{vJ}(\xi), \quad [2]$$

with

$$\mu_c^{-1} = (M_a + M_b - Cm_e)/M_aM_b. \quad [3]$$

The mass-independent functions $Q_{a,b}(\xi)$, $R_{a,b}(\xi)$, and $S_{a,b}(\xi)$ are the correction terms for the nonadiabatic vibrational, nonadiabatic rotational, and adiabatic effects, respectively (*5*). The quantities M_a, M_b, and m_e are the atomic masses of atoms a and b and the electron mass, respectively. C is the charge number of the molecule (*2*).

According to a procedure similar to that described by Herman and Asgharian (*6*) Eq. [2] is altered to a form amenable to a WKB treatment. With a substitution

$$\Psi_{vJ}(\xi) = \psi_{vJ}(\xi)\exp[-(1/2)\{(m_e/M_a)Q_a(\xi) + (m_e/M_b)Q_b(\xi)\}], \quad [4]$$

and neglecting terms of orders higher than $O(m_e/M_i)$, the equation

$$\left[-\frac{h^2}{8\pi^2\mu_c r_e^2}\{1 + (m_e/M_a)Q_a(\xi) + (m_e/M_b)Q_b(\xi)\}\frac{d^2}{d\xi^2} + \frac{h^2}{16\pi^2\mu_c r_e^2}\{(m_e/M_a)Q_a''(\xi) + (m_e/M_b)Q_b''(\xi)\} + \frac{h^2}{8\pi^2\mu_c r_e^2(1+\xi)^2} \times \{1 + (m_e/M_a)R_a(\xi) + (m_e/M_b)R_b(\xi)\}J(J+1) + V(\xi) + (m_e/M_a)S_a(\xi) + (m_e/M_b)S_b(\xi)\right]\psi_{vJ}(\xi) = E_{vJ}\psi_{vJ}(\xi) \quad [5]$$

is obtained, in which $''$ denotes second derivative. As the terms $(h^2/16\pi^2\mu_c r_e^2)\{(m_e/M_a)Q_a''(\xi) + (m_e/M_b)Q_b''(\xi)\}$ are smaller than the $(m_e/M_{a,b})S_{a,b}(\xi)$ terms (*6*), they can be left out.

The Dunham potential energy function $V(\xi)$ (*9*) is expressed as

$$V(\xi) = (1/2)kr_e^2\xi^2(1 + a_1\xi + a_2\xi^2 + \cdots). \quad [6]$$

Similarly, after Ogilvie (*5*), functions $Q_{a,b}(\xi)$, $R_{a,b}(\xi)$, and $S_{a,b}(\xi)$ are expressed as Taylor expansions of ξ as

$$Q_{a,b}(\xi) = \sum_{i=0} q_i^{a,b}\xi^i, \quad [7]$$

$$R_{a,b}(\xi) = \sum_{i=0} r_i^{a,b}\xi^i, \quad [8]$$

$$S_{a,b}(\xi) = \sum_{i=1} s_i^{a,b}\xi^i, \quad [9]$$

in which $s_0^{a,b}$ terms are removed from $S_{a,b}(\xi)$ because they yield no effects on Eq. [5]. The notations $r_i^{a,b}$ for the expansion coefficients are discriminated from the notations r and r_e for internuclear distances by attaching the subscript i. The expansion coefficients $q_i^{a,b}$, $r_i^{a,b}$, and $s_i^{a,b}$ correspond to Ogilvie's coefficients $s_j^{a,b}$, $t_j^{a,b}$, and $u_j^{a,b}$, respectively, though he expanded the functions with the normalized internuclear distance z [$=2(r - r_e)/(r + r_e)$].

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Dividing Eq. [5] by

$$1 + (m_e/M_a)\sum_{i=1} q_i^a \xi^i + (m_e/M_b)\sum_{i=1} q_i^b \xi^i$$

we obtain

$$\left[-\frac{h^2}{8\pi^2\mu_c r_e^2}(1+\delta q_0)\frac{d^2}{d\xi^2} + \frac{h^2}{8\pi^2\mu_c r_e^2(1+\xi)^2} \right.$$
$$\times(1+\delta r_0)(1+\sum_{i=1}\delta r_i'\xi^i)J(J+1) \quad [10]$$
$$+ (1/2)kr_e^2\xi^2\{1+\sum_{i=1}(a_i-\delta q_i)\xi^i\}$$
$$\left. + \sum_{i=1}\delta s_i'\xi^i \right]\psi_{vJ}(\xi) = E_{vJ}\psi_{vJ}(\xi),$$

where

$$\delta q_i = (m_e/M_a)q_i^a + (m_e/M_b)q_i^b, \quad [11]$$
$$\delta r_0 = (m_e/M_a)r_0^a + (m_e/M_b)r_0^b, \quad [12]$$
$$\delta r_i' = (m_e/M_a)(r_i^a - q_i^a) + (m_e/M_b)(r_i^b - q_i^b), \quad [13]$$

and

$$\delta s_i' = (m_e/M_a)(s_i^a + q_i^a E_{vJ}) + (m_e/M_b)(s_i^b + q_i^b E_{vJ}). \quad [14]$$

We can calculate the energy values by applying the WKB treatment to Eq. [10]. The calculation is given in the Appendix. The quantity E_{vJ} that appears on the left side of Eq. [10] can be expressed as a known power series expansion in $(v + 1/2)$ and $J(J+1)$.

However, for the present purpose of obtaining analytic expressions for Δ_ω and Δ_B algebraically, further reduction of Eq. [10] is necessary. We assume that the minimum of the effective potential function that consists of $V(\xi)$ plus the adiabatic correction occurs at an internuclear distance r_e'. We introduce a variable $\eta = (r - r_e')/r_e'$ so that the variable ξ becomes converted to η by the relation

$$\xi = (1+t)\eta + t, \quad [15]$$
$$t = (r_e' - r_e)/r_e$$
$$= -(1/kr_e^2)\delta s_1'. \quad [16]$$

t is determined in order to remove from the potential function the term $\delta s_1'\xi$, which is linear in ξ.

The Schrödinger equation [10] becomes rewritten as

$$\left[-\frac{h^2(1+g_1)}{8\pi^2\mu_c r_e^2}\frac{d^2}{d\eta^2} + \frac{h^2(1+g_2)}{8\pi^2\mu_c r_e^2(1+\eta)^2} \right.$$
$$\times(1+\sum_{i=1}\delta r_i'\eta^i)J(J+1) \quad [17]$$
$$\left. + (1/2)kr_e^2(1+g_3)\eta^2(1+\sum_{i=1}k_i\eta^i) \right]\psi_{vJ}(\eta)$$
$$= E_{vJ}\psi_{vJ}(\eta),$$

in which

$$g_1 = \delta q_0 + (2/kr_e^2)\delta s_1', \quad [18]$$
$$g_2 = \delta r_0 + (2/kr_e^2)\delta s_1', \quad [19]$$
$$g_3 = -(1/kr_e^2)\{(2+3a_1)\delta s_1' - 2\delta s_2'\}, \quad [20]$$
$$k_1 = a_1 - \delta q_1 - (1/kr_e^2)\{(a_1 - 3a_1^2 + 4a_2)\delta s_1' + 2a_1\delta s_2' - 2\delta s_3'\}, \quad [21]$$

and

$$k_2 = a_2 - \delta q_2 - (1/kr_e^2)\{(2a_2 - 3a_1a_2 + 5a_3)\delta s_1' + 2a_2\delta s_2' - 2\delta s_4'\}, \text{ etc.} \quad [22]$$

CALCULATION OF ENERGY VALUES

Before we proceed to the calculations, Dunham's process (*9, 21*) is briefly reviewed. Dunham calculated the characteristic values of the Schrödinger equation

$$\frac{d^2\psi}{d\xi^2} + \frac{8\pi^2\mu r_e^2}{h^2}\left[E - V - \frac{h^2 J(J+1)}{8\pi^2\mu r_e^2(1+\xi)^2} \right]\psi = 0, \quad [23]$$

in which V is given by Eq. [6]. Replacing E and V by hcF and hcU, respectively, functions of U were given as

$$U' = \frac{dU}{d\xi} = A_1U^{1/2} + A_2U + A_3U^{3/2} + A_4U^2 + \cdots, \quad [24]$$

and

$$(U')^{-1} = (A_1U^{1/2})^{-1}[1 + B_1U^{1/2} + B_2U + B_3U^{3/2} + \cdots]. \quad [25]$$

The expansion coefficients A_1, A_2, $\cdots$ and B_1, B_2, $\cdots$ can be expressed in terms of a_0, a_1, a_2, $\cdots$. After the

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rotational energy term in Eq. [23] was absorbed into U, the energy values F_{vJ} were obtained by the equation[1]

$$\begin{aligned} F &= B_e A_1 A_3/16 + A_1 B_e^{1/2}(v+1/2) + (B_e^{3/2}A_1^2/32) \\ &\times (5A_5 - B_2A_3)(v+1/2) - (B_eB_2A_1^2/4) \\ &\times (v+1/2)^2 + (B_e^2A_1^3/128)(35A_7 - 15B_2A_5 \\ &+ 3A_3B_2^2 - 3A_3B_4)(v+1/2)^2 + (B_e^{3/2}A_1^3/8) \\ &\times (B_2^2 - B_4)(v+1/2)^3 + (B_e^2A_1^4/64)(10B_2B_4 \\ &- 5B_2^3 - 5B_6)(v+1/2)^4 + \cdots, \end{aligned} \quad [26]$$

which was derived by the WKB treatment. The resulting F_{vJ} was expressed as

$$F_{vJ} = \sum_{ij} Y_{ij}(v+1/2)^i[J(J+1)]^j, \quad [27]$$

in which Dunham's coefficient Y_{ij} consists of contributions in a series

$$Y_{ij} = Y_{ij}^{(0)} + Y_{ij}^{(2)} + Y_{ij}^{(4)} + \cdots. \quad [28]$$

The superscripts indicate the order of the integral in the WKB treatment and each successive contribution $Y_{ij}^{(n)}$ decreases proportionally to a factor $(4B_e^2/\omega_e^2)$.

In the present case, all quantities involving the energy in Eq. [17] are similarly expressed in wavenumber units; for example, V is replaced by hcU and also for $\delta s_i'$ and E_{vJ}. Taking into account the rotational energy, the potential function U_J is given as

$$\begin{aligned} U_J &= \kappa_0\eta^2(1 + k_1\eta + k_2\eta^2 + \cdots) \\ &+ B_e^*J(J+1)\{1 - 2(1 - \delta r_1'/2)\eta \\ &+ 3(1 - 2\delta r_1'/3 + \delta r_2'/3)\eta^2 - \cdots\}, \end{aligned} \quad [29]$$

with

$$\kappa_0 = \omega_e^{*2}/4B_e^* \quad [30]$$

$$\begin{aligned} B_e^* &= B_e\{1 + \delta r_0 + (4B_e/\omega_e^2)\delta s_1'\} \\ &= B_e[1 + (m_e/M_a)\{\Delta_B^a + (4B_e/\omega_e^2)q_1^aF_{vJ}\} \\ &+ (m_e/M_b)\{\Delta_B^b + (4B_e/\omega_e^2)q_1^bF_{vJ}\}] \end{aligned} \quad [31]$$

[1] There is a misprint in Eq. [12] of Ref. (9); the sixth term on the right-hand side, $(B_e^{3/2}A_1^3/8)(B_2^2 - B_4)(v+1/2)^2$, should read $(B_e^{3/2}A_1^3/8)(B_2^2 - B_4)(v+1/2)^3$.

$$\begin{aligned} \omega_e^* &= \omega_e\{1 + \delta r_0/2 - (B_e/\omega_e^2)(3a_1\delta s_1' - 2\delta s_2')\} \\ &= \omega_e[1 + (m_e/M_a)\{\Delta_\omega^{ra} - (B_e/\omega_e^2)(3a_1q_1^a - 2q_2^a)F_{vJ}\} \\ &+ (m_e/M_b)\{\Delta_\omega^{rb} - (B_e/\omega_e^2)(3a_1q_1^b - 2q_2^b)F_{vJ}\}] \end{aligned} \quad [32]$$

and

$$\begin{aligned} k_1 &= a_1[1 - \delta q_1/a_1 - (2B_e/\omega_e^2)\{(1 - 3a_1 + 4a_2/a_1) \\ &\times \delta s_1' + 2\delta s_2' - 2\delta s_3'/a_1\}] \\ &= a_1(1 + (m_e/M_a)[\Delta_{a1}^a - (2B_e/\omega_e^2)\{(1 - 3a_1 \\ &+ 4a_2/a_1)q_1^a + 2q_2^a - 2q_3^a/a_1\}F_{vJ}] + (m_e/M_b) \\ &\times [\Delta_{a1}^b - (2B_e/\omega_e^2)\{(1 - 3a_1 + 4a_2/a_1)q_1^b \\ &+ 2q_2^b - 2q_3^b/a_1\}F_{vJ}]), \text{ etc.} \end{aligned} \quad [33]$$

Again we introduce a variable ρ such that $\eta = \rho + \epsilon$ and choose ϵ so that the minimum of U_J falls at the point $\rho = 0$. Then U_J is expressed as a power series in ρ of the same form as Eq. [6]; the energy values F_{vJ} are obtained using Eq. [26] as

$$F_{vJ} = \sum_{ij} Y_{ij}^{*vJ}(v+1/2)^i[J(J+1)]^j, \quad [34]$$

in which we use the notation Y_{ij}^{*vJ} to include the contributions from Born–Oppenheimer correction terms to distinguish from Dunham's original notation Y_{ij}. The superscript vJ indicates that the Y_{ij}^{*vJ} coefficient includes the quantity F_{vJ} that is expressed as the polynomial of $(v+1/2)^i[J(J+1)]^j$.

Six Y_{ij}^{*vJ} selected for an analysis of LiH spectra are

$$Y_{01}^{*vJ} = B_e^* + Y_{01}^{*vJ(2)} + \cdots, \quad [35]$$

$$Y_{02}^{*vJ} = -(4B_e^{*3}/\omega_e^{*2})(1 - \delta r_1') + Y_{02}^{*vJ(2)} + \cdots, \quad [36]$$

$$\begin{aligned} Y_{03}^{*vJ} &= (16B_e^{*5}/\omega_e^{*4})\{3 + k_1 - (5 + 3k_1/2)\delta r_1' + \delta r_2'\} \\ &+ Y_{03}^{*vJ(2)} + \cdots, \end{aligned} \quad [37]$$

$$Y_{10}^{*vJ} = \omega_e^*(1 - \delta r_0'/2) + Y_{10}^{*vJ(2)} + \cdots, \quad [38]$$

$$\begin{aligned} Y_{11}^{*vJ} &= (B_e^{*2}/\omega_e^*)\{6(1 + k_1) - 3(1 + k_1)\delta r_0' \\ &- (4 + 3k_1)\delta r_1' + 2\delta r_2'\} + Y_{11}^{*vJ(2)} + \cdots, \end{aligned} \quad [39]$$

and

$$\begin{aligned} Y_{20}^{*vJ} &= -(3B_e^*/8)(5k_1^2 - 4k_2)(1 - \delta r_0') \\ &+ Y_{20}^{*vJ(2)} + \cdots. \end{aligned} \quad [40]$$

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In Eqs. [35]–[40] we show results of only the lowest-order terms $Y_{ij}^{*vJ(0)}$ because effects of Born–Oppenheimer corrections are negligible for the $Y_{ij}^{*vJ(n)}$ coefficients with $n \geqslant 2$. If more $Y_{ij}^{*vJ(0)}$ other than those given above are required, it is easy to calculate their expressions according to the present procedure. The higher-order contributions $Y_{ij}^{*vJ(n)}$ ($n \geqslant 2$) in Eqs. [35]–[40] are assumed to have the same form as Dunham's $Y_{ij}^{(n)}$ ($n \geqslant 2$) except that B_e, ω_e, a_1, a_2, $\cdots$ in $Y_{ij}^{(n)}$ ($n \geqslant 2$) become replaced by B_e^*, ω_e^*, k_1, k_2, $\cdots$, respectively, as the present Y_{ij}^{*vJ} are derived from an effective potential function Eq. [29]. Y_{ij}^{*vJ} other than those of Eqs. [35]–[40] are assumed to have the same form as the corresponding Dunham's Y_{ij}.

APPLICATION TO LiH

In order to test the present model including adiabatic and nonadiabatic corrections we analyzed a data set of lithium hydride consisting of vibrational–rotational and rotational spectra reported by Maki *et al.* (*15*) and microwave spectra from Plummer *et al.* (*22*) and from Pearson and Gordy (*23*). Although Bellini *et al.* (*20*) reported an additional 22 rotational transitions for lithium hydride, we did not include them because we intended to test the present model by comparing the quality of the fit with work of Maki *et al.* (*15*) and of Tiemann and Ogilvie (*4*). The results of Tiemann and Ogilvie (*4*) and of preceding reports by Ogilvie (*8, 16*) are the first in which analysis of the spectra of lithium hydride included a correction as the perturbation treatment for a Born–Oppenheimer approximation unambiguously with an algebraic approach based on a Dunham formalism.

The data set we chose was the same as that of Tiemann and Ogilvie (*4*) consisting of 151 rotational and 354 vibrational–rotational FTS (Fourier transform spectrometer) measurements (*15*), 27 vibrational–rotational diode laser measurements (*15*), and 10 millimeter-wave measurements (*22, 23*). Excluding one transition $R(8)$ of the band ($v = 3$–2) of ^{7}Li^{1}H after Ogilvie (*8*), in total 541 spectral lines for four isotopomers of lithium hydride were analyzed using Eqs. [35]–[40]. Other Y_{ij} and higher-order contributions for Eqs. [35]–[40] were taken from Bouanich (*24*), Woolley (*25*), Ogilvie and Bouanich (*26*), and Ogilvie and Tipping (*27*), although B_e, ω_e, a_1, a_2, $\cdots$ in those Y_{ij} were replaced by B_e^*, ω_e^*, k_1, k_2, $\cdots$, respectively.

The transitions for four isotopomers were fitted simultaneously to the potential model retaining 19 adjustable parameters, U_ω, U_B, a_1, a_2, a_3, a_4, a_5, a_6, a_7, Δ_ω^H, Δ_ω^{Li}, q_1^H, Δ_B^H, Δ_B^{Li}, $r_0'^H$, $r_1'^H$, $r_1'^{Li}$, $r_2'^H$, and Δ_{a1}^H, in which

$$U_\omega = \omega_e \mu_c^{1/2}, \qquad [41]$$

$$U_B = B_e \mu_c, \qquad [42]$$

$$\Delta_\omega^{a,b} = \Delta_\omega^{ra,b} - r_0'^{a,b}/2, \qquad [43]$$

and

$$r_i'^{a,b} = r_i^{a,b} - q_i^{a,b}. \qquad [44]$$

Other parameters were set to zero. These parameters were connected with the vibrational–rotational energy levels through 28 Y_{ij}^{*vJ}, i.e., ij = 10, 20, 30, 40, 50, 01, 11, 21, 31, 41, 02, 12, 22, 32, 42, 03, 13, 23, 33, 04, 14, 24, 05, 15, 06, 16, 07, and 08. Other higher-order constants were indeterminate and set equal to zero according to our previous procedure (*12*). Atomic masses were taken from Audi and Wapstra (*28*).

The weights for observed data were assumed to be proportional to $(1/\delta_{obs})^2$. After Tiemann and Ogilvie (*4*), we assigned errors of measurements to be 0.0008, 0.0006, and 1.7×10^{-6} cm^{-1} for the vibrational–rotational transitions, the rotational transitions of the far-infrared data, and the millimeter-wave transitions, respectively. With 19 parameters the fit converged after three iterations. The normalized standard deviation was 1.07, which corresponded to a standard deviation of 0.00086 cm^{-1} for IR transitions and is comparable with a root-mean-square deviation of 0.00096 cm^{-1} obtained by Maki *et al.* (*15*) in a fit with empirical parameters U_{ij} and Δ_{ij}. The reduced standard deviations reported by Tiemann and Ogilvie (*4*) were 1.06 for an empirical U_{ij} and Δ_{ij} fit and 1.10 for a fit with an algebraic approach. In the present fit seven parameters for Born–Oppenheimer corrections were determined for hydrogen in contrast with the case of Tiemann and Ogilvie in which six Δ_{ij} were determined for hydrogen. The reason is that the parameters Δ_ω^H, q_1^H, Δ_B^H, and Δ_{a1}^H in the present analysis are included not only in the corresponding six Y_{ij}^{*vJ} but also all relevant Y_{ij}^{*vJ}. Therefore, the standard deviation indicates that the quality of the present fit is satisfactory. The molecular parameters determined in the present fit are shown in Table 1 with the previous results by Maki *et al.* (*15*) of their potential model (*11*).

DISCUSSION

The contributions from the Born–Oppenheimer corrections are represented by compact analytic expressions of Eqs. [35]–[40]. Using this modification of the model of Thompson *et al.* (*11*) we are able to fit the vibrational–rotational and rotational spectra of four isotopomers of LiH simultaneously with a single set of molecular parameters within estimated experimental errors. The rotational parameter $B_e = h/(8\pi^2 \mu_c r_e^2 c)$ and the vibrational parameter $\omega_e = (1/2\pi c)(k/\mu_c)^{1/2}$, which are molecular parameters within the Born–Oppenheimer approximation, are extracted with Eqs. [41] and [42].

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TABLE 1
Dunham Potential Constants for Lithium Hydride

Parameters	This work	Ref. (15)
U_ω /cm^{-1}u$^{1/2}$	1319.941 5(15) [a]	1319.986 8(57)
Δ_ω^{Li}	−0.112 3(57)	−0.164(33)
Δ_ω^{H}	−0.667 0(16)	−0.746 4(18)
q_1^{H}	−0.60(40)	
U_B /cm^{-1}u^{1}	6.627 096 5(44)	6.627 153(24)
Δ_B^{Li}	−0.113 6(74)	−0.136(40)
Δ_B^{H}	−1.555 77(66)	−1.578 2(20)
$r_0'^{H}$	−1.09(79)	
$r_1'^{Li}$	0.41(13)	
$r_1'^{H}$	2.51(40)	
$r_2'^{H}$	−2.07(29)	
a_1	−1.897 154(24)	−1.897 234(64)
Δ_{a1}^{H}	−0.188(27)	
a_2	2.444 09(12)	2.441 78(56)
a_3	−2.635 5(14)	−2.633 4(27)
a_4	2.500 3(36)	2.584 2(88)
a_5	−2.082(21)	−2.670(62)
a_6	1.33(11)	3.43(24)
a_7	−0.34(24)	−4.55(48)

[a] The uncertainty (one standard error) in the last digits is given in parentheses.

Table 1 shows that the dominant part of the contributions from the adiabatic and nonadiabatic effects is absorbed in $\Delta_\omega^{H,Li}$ and $\Delta_B^{H,Li}$. Moreover, the contributions to Y_{10}^{vJ} and Y_{01}^{vJ} coefficients are well represented with $\Delta_\omega^{H,Li}$ and $\Delta_B^{H,Li}$, respectively. Therefore, for massive molecules for which we can detect adiabatic and nonadiabatic effects only on Y_{10} and Y_{01}, those effects are described well with two parameters $\Delta_\omega^{a,b}$ and $\Delta_B^{a,b}$ as shown previously. This result gives a basis for the validity of the model of Thompson *et al.* (*11*) for the analysis of heavy molecules, because small effects of F_{vJ} terms can be neglected.

It should be noted that the parameter a_7 was not determined significantly in the present analysis because we did not include in the analysis some of the Y_{ij}^{vJ} coefficients which were relevant in determining a_7.

From Eqs. [35] and [38] we see $U_\omega = U_{10} = U_{10}^{(1)}$ and $U_B = U_{01} = U_{01}^{(1)}$, in which U_{10} and U_{01} are those given in Refs. (*15, 19*) and $U_{10}^{(1)}$ and $U_{01}^{(1)}$ are given in Ref. (*4*). A comparison of values of these parameters appears in Table

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TABLE 2
Reduced Dunham Parameters U_{10} and U^{a}_{01}

Sources	$10^{-3}\,U_{10}$ /cm^{-1} u$^{1/2}$	U_{01} /cm^{-1} u^{1}
This work (Potential fit)	1.319 941 5(15) [b]	6.627 096 5(44)
This work (U_{ij}, Δ_{ij} fit)	1.319 946 9(12)	6.627 101 6(44)
Maki et al. (15) (U_{ij}, Δ_{ij} fit)	1.319 948 8(13)	6.627 104 7(46)
Ogilvie (19) (U_{ij}, Δ_{ij} fit)	1.319 947 0(12)	6.627 103 7(10)
Ogilvie (19) (Potential fit)	1.319 947 5(11)	6.627 104 4(11)

[a] See Discussion in the text.
[b] The uncertainty (one standard error) in the last digits is given in parentheses.

2. Ogilvie (*19*) revised results of Ref. (*4*) by using correct masses (*28*) and an extended data set of lithium hydride which included all available data (*20, 29, 30*) other than those used in Ref. (*4*). We list only the revised values (*19*) in Table 2.

Our value of U_ω of the present fit (potential fit) differs slightly from corresponding values of other sources listed in rows 3 to 5 of Table 2. To explore the reason for the slight discrepancy of our value of U_ω, we fitted our data set to freely floating U_{ij} and Δ_{ij} parameters partially imposing constraints according to Maki *et al.* (*15*). With values of U_{32}, U_{23}, U_{14}, U_{15}, U_{06}, U_{16}, U_{07}, and U_{08} constrained to those of Ref. (*4*), the 24 parameters, i.e., U_{10}, U_{20}, U_{30}, U_{40}, U_{01}, U_{11}, U_{21}, U_{31}, U_{02}, U_{12}, U_{22}, U_{03}, U_{13}, U_{04}, U_{05}, Δ^{H}_{10}, Δ^{H}_{20}, Δ^{H}_{01}, Δ^{H}_{11}, Δ^{H}_{02}, Δ^{H}_{03}, Δ^{Li}_{10}, Δ^{Li}_{01}, and Δ^{Li}_{02}, were determined using Eq. [1] where values of U_{50}, U_{41}, U_{42}, U_{33}, and U_{24} were set to zero in the fit. The values of U_{10} and U_{01} thus obtained are listed in Table 2 in row 2 (U_{ij}, Δ_{ij} fit). Our values of U_{10} for the two cases (potential fit and U_{ij}, Δ_{ij} fit) overlap mutually within 3σ; hence we conclude that the present model of Eqs. [35]–[40] works well for lithium hydride.

We can rearrange the right side of Eq. [34] to extract the F_{vJ} terms from the Y^{*vJ}_{ij} coefficients and to expand them in a $(v + 1/2)^i[J(J + 1)]^j$ series. The rearranged series expansion has term coefficients Y^{*}_{ij} that lack vJ-dependent terms. Six Y^{*}_{ij} corresponding to Eqs. [35]–[40] are

$$Y^{*(0)}_{01} = B^{d}_{e}, \qquad [45]$$

$$Y^{*(0)}_{02} = -(4B^{d3}_{e}/\omega^{d2}_{e})(1 - \delta r_1), \qquad [46]$$

$$Y^{*(0)}_{03} = (16B^{d5}_{e}/\omega^{d4}_{e})\{3 + a^{d}_{1} - (5 + 3a_1/2)\delta r_1 + \delta r_2\}, \qquad [47]$$

$$Y^{*(0)}_{10} = \omega^{d}_{e}(1 - \delta r'_0/2), \qquad [48]$$

$$Y^{*(0)}_{11} = (B^{d2}_{e}/\omega^{d}_{e})\{6(1 + a^{d}_{1}) - 3(1 + a_1)\delta r'_0 - (4 + 3a_1)\delta r_1 + 2\delta r_2 + 2\delta q_1\}, \qquad [49]$$

and

$$Y^{*(0)}_{20} = -(B^{d}_{e}/8)\{3(5a^{d2}_{1} - 4a^{d}_{2}) - 3(5a^{2}_{1} - 4a_2)\delta r'_0 - 6a_1\delta q_1 - 4\delta q_2\}, \qquad [50]$$

in which

$$B^{d}_{e} = B_e[1 + (m_e/M_a)\Delta^{a}_{B} + (m_e/M_b)\Delta^{b}_{B}], \qquad [51]$$

$$\omega^{d}_{e} = \omega_e[1 + (m_e/M_a)\Delta^{ra}_{\omega} + (m_e/M_b)\Delta^{rb}_{\omega}], \qquad [52]$$

and

$$a^{d}_{1} = a_1[1 + (m_e/M_a)\Delta^{a}_{a1} + (m_e/M_b)\Delta^{b}_{a1}]. \qquad [53]$$

These Y^{*}_{ij} coefficients of Eqs. [45]–[50] that exclude vJ-dependent terms are consistent with rovibrational energy Eq. [1].

In conclusion, we have proposed a potential model that consists of compact expressions and is an extension of the potential model by Thompson *et al.* (*11*). The model is applicable to light molecules such as LiH that show signifi-

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cant contributions of adiabatic and nonadiabatic effects. The analytical expressions derived for $\Delta_{\omega}^{a,b}$ and $\Delta_{B}^{a,b}$ in this work provide the basis for the validity of the potential model of Thompson *et al.* (*11*) for an analysis of molecules composed of heavier atoms.

APPENDIX

We introduce a variable ζ such that $\xi = \zeta + \epsilon(\Lambda, s)$, in which $\Lambda = J(J+1)$ and $s = \delta s_1'$, and choose ϵ so that terms linear in ξ on the left side of Eq. [10] are removed. Then we express the potential as a power series in ζ corresponding to Eq. [6]. The energy values are obtained by substituting the potential coefficients in Eq. [26]. After all quantities involving the energy are converted to wavenumber units, the results are the same in this order of approximation as the expressions given in Eqs. [35]–[40]; we show only selected examples of $Y_{01}^{*vJ(0)}$, $Y_{02}^{*vJ(0)}$, and $Y_{10}^{*vJ(0)}$:

$$Y_{01}^{*vJ(0)} = B_e\{(1 + \delta r_0) + (4B_e/\omega_e^2)\delta s_1'\}, \qquad [54]$$

$$Y_{02}^{*vJ(0)} = -(4B_e^3/\omega_e^2)[1 + 2\delta r_0 - \delta r_1' + (2B_e/\omega_e^2)\{(6 + 3a_1)\delta s_1' - 2\delta s_2'\}], \qquad [55]$$

and

$$Y_{10}^{*vJ(0)} = \omega_e\{1 + \delta q_0/2 - (2B_e/\omega_e^2)(3a_1\delta s_1'/2 - \delta s_2')\}. \qquad [56]$$

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